

Applied Databases Project – Innovation

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1. Attendee Connection Analytics (Neo4j) - Improvement

An enhancement was made to the “View Connected Attendees” functionality to include a count of the number of connections each attendee has.

Example Output

Attendee 106 has 2 connections:

- Attendee 103
- Attendee 114

This provides additional insight into the dataset and improves readability by summarising relationship data, rather than only listing connections.

2. Total Attendees per Company - Improvement

The “View Attendees by Company” option was extended to display the total number of attendees belonging to a company. This ensures accurate counting by removing duplicate entries where attendees attend multiple sessions. A Python set() was used to ensure only unique attendee names were counted.

Example Output

Company: CloudSprint

Total Attendees: 3

This adds a useful summary statistic that helps users quickly understand company participation levels.

3. View Sessions by Date – Additional Feature

An additional menu option was implemented to allow users to view all sessions taking place on a selected date. This option:

- Prompts the user to enter a session date
- Displays session title, speaker, and room location
- Uses a SQL join between the session and room tables
- Input is validated using Python’s datetime module to ensure correct date format (YYYY-MM-DD)

This improves usability by allowing users to navigate the schedule based on which day they will be attending, which is a common real-world requirement.

4. Check Registration – Additional Feature

A new feature was implemented to allow users to view all sessions that a specific attendee is registered to attend. This option:

- Prompts the user to enter an attendee ID
- Retrieves all sessions linked to that attendee via the registration table
- Displays session title, speaker, date, and location
- Checks that the attendee ID is numeric
- Confirms the attendee exists in the database

This provides a practical and user-focused feature that allows individuals to quickly verify their schedule, demonstrating understanding of relationships between multiple database tables.

The above innovations demonstrate the ability to extend the base system with meaningful enhancements. My additions improve usability, provide additional data insights, and show a deeper understanding of both MySQL and Neo4j database systems, as well as Python application design.